Christian Garry

Quantitative Researcher — stochastic control · pricing models · machine learning
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Profile

Quantitative-finance researcher (MSc Scientific Computing, Distinction track ∼86%) who builds and validates **models, strategies, and systems for pricing and trading**. My dissertation develops **deep solvers for a competitive market-making game** — pricing and quoting under inventory risk and mean-field competition — validated to **0.135%** against an exact benchmark, and studies how **trading-strategy learning** behaves via multi-agent reinforcement learning, with a reproducible, statistically-significant finding on equilibrium vs. learned spreads. A strong Python programmer with deep grounding in **stochastic control, probability/statistics, optimisation, and time-series**, a genuine research track record (a 10-chapter dissertation + three standalone papers), and a precise, honest communication style.

Research

Deep BSDEs for Mean-Field Market Making — MSc Dissertation, Durham University · *2026*

Quantitative finance × stochastic control × machine learning · github.com/cgarryZA/MFG-BSDE-Equilibrium

- Built **models for pricing and quoting** in the Cont–Xiong dealer market-making game — intensity-controlled fills, inventory risk, mean-field competition — using deep backward-SDE-with-jumps solvers; **validated to 0.135% spread / 0.216% value error** against an exact benchmark and **scaled to ∼5,000 agents** (mean-field rate $O(1/\sqrt{N})$ confirmed) where grid/PDE methods are infeasible.
- **Studied trading-strategy learning dynamics:** trained **20 seeded multi-agent RL (decentralised MADDPG)** agents on the same game → population mean spread **1.866** (95% CI [1.654, 2.072]), above both the competitive Nash anchor (1.515) and the cartel level (1.593); **t=3.22, p=0.0022**, Cohen's d=0.72, surviving Bonferroni & Holm correction. Established that the game has a *unique* competitive equilibrium with no supra-cartel fixed point — so the effect is a property of the **learning algorithm**, not the market (direct algorithmic-collusion / surveillance relevance: FCA, PRA, SR 11-7).
- **Feature/architecture engineering for financial-dataset models:** identified and fixed three failure modes (generator bypass, BatchNorm erasure, DeepSets collapse) that silently destroy the population/competition signal in mean-field networks; validated across a **26-experiment** battery (post-fix, learned competition factor varies 4× and optimal quotes shift 2×).
- Developed an **a-posteriori error-certification theory** (a computable estimator with two-sided guarantees and a certified stopping rule) and machine-checked its core in **Lean 4 / Mathlib** (∼36k LOC, axiom-clean) — rigour that carries straight into trusting a model's numbers.
- Built **reproducible-research infrastructure**: a CI-guarded data→results pipeline (build fails if any cited number drifts), a GPU job queue, and remote automated job submission.

Silicon Carbide JFET CPU — MEng Dissertation, Durham University · *2023–24 · 84%*

- Built a custom 4-bit CPU and its **full verification toolchain** (compiler, assembler, automation, emulator-parity + waveform-to-state checks) — instruction-level validation against a behavioural reference. Demonstrates end-to-end engineering rigour and modelling-to-validation discipline.

Education

MSc Scientific Computing & Data Analysis (AI for Engineering) — Durham University · *Sep 2025 – Sep 2026*

- Tracking **Distinction** (∼86%). Directly relevant coursework: **Bayesian inference & statistics** (MCMC, Gaussian processes — incl. a high-dimensional gene-expression *survival regression* and **time-series / regression** modelling), **convex optimisation** (duality, KKT) applied to estimation & SVMs, **random forests/boosting & deep learning**, and **GPU/HPC performance engineering** (for large financial datasets).
- Standout marks: **95%** Performance/GPU, **94%** Machine Learning & Statistics, **∼89%** in both Advanced Bayesian ML modules. Python, C, R.

Mathematics modules (NetMath) — University of Illinois Urbana-Champaign · *2026*

- Credit-bearing, exam-assessed proof-based mathematics taken alongside the MSc: **MATH 314** Introduction to Higher Mathematics (in progress, current average 100%) and **MATH 447** Real Analysis (upcoming) —

sequences and convergence, metric spaces, compactness, and Riemann integration. Strengthens the rigorous-analysis foundations under stochastic calculus and probability.

MEng Electronic Engineering — Durham University · *2020–2024* · *Upper Second-Class Honours (2:1)*

Experience

Graduate Communications Engineer — Siemens PLC · *Sep 2024 – Present*

- Designed and built a **Retrieval-Augmented Generation platform from scratch in Python** (local LLMs + Qdrant vector search) over large technical corpora **— cutting engineer time-to-answer by >90%**, with transparent, reproducible behaviour.
- Implemented **C/C++ data-exchange / networking components** (TCP/IP, serial) for digital-twin simulation environments.

Technical Skills

Quantitative / stochastic: BSDEs/BSDEJs, McKean–Vlasov & mean-field games, HJB↔BSDE stochastic control, Monte-Carlo & finite-difference numerics, market microstructure / optimal market making, inventory risk, fixed-point/equilibrium computation.

Statistics / ML: Bayesian inference (MCMC, Gaussian processes), time-series & regression modelling, feature engineering, random forests/boosting, deep learning (PyTorch), multi-agent RL (MADDPG), hypothesis testing with multiple-comparison correction, bootstrap CIs.

Engineering: Python (strong), C++, C, R; GPU/CUDA & HPC (roofline, vectorisation, Slurm); Git/CI, reproducible-research pipelines; Lean 4 / Mathlib.